

Decision under Uncertainty and Portfolio Analysis

Lecture 3: IRFA and MMMEF

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Alternatives to Expected Utility

The Allais and common-ratio effects suggest that a common lottery component can change the ranking of the remaining risks.

The restriction we reconsider: Independence

For all lotteries P, Q, R and $\lambda \in (0, 1)$,

$$P \succeq Q \iff \lambda P + (1 - \lambda)R \succeq \lambda Q + (1 - \lambda)R.$$

- ▶ **Betweenness models:** retain neutrality toward mixing indifferent lotteries.
- ▶ **Local EU:** retain smooth local evaluation, without global linearity.
- ▶ **Cautious EU:** preserve a one-sided comparison with certainty.

Chew–Dekel Preferences: Betweenness

Replacement axiom: Betweenness

For all P, Q and $\lambda \in (0, 1)$,

$$P \sim Q \implies P \sim \lambda P + (1 - \lambda)Q \sim Q,$$

$$P \succ Q \implies P \succ \lambda P + (1 - \lambda)Q \succ Q.$$

- ▶ Randomization between indifferent lotteries has no value or cost.
- ▶ Mixing a better and a worse lottery gives an intermediate option.
- ▶ Independence implies betweenness. Betweenness does not require comparisons to survive mixing with an arbitrary third lottery R .

Behavioral implication

Indifference lines in the probability triangle are straight, but can have different slopes. Common-consequence reversals are therefore possible.

Chew–Dekel Preferences: Implicit Evaluation

Under the usual ordering, continuity, and regularity conditions, betweenness preferences admit an implicit representation:

$$V(P) = \sum_i p_i \Gamma(x_i, V(P)).$$

- ▶ The contribution of outcome x_i can depend on the overall value $V(P)$.
- ▶ At a fixed value v , indifference satisfies

$$\sum_i p_i [\Gamma(x_i, v) - v] = 0,$$

which is linear in probabilities.

- ▶ Different values v can generate different slopes: lines need not be parallel.
- ▶ If $\Gamma(x, v) = u(x)$, the equation becomes ordinary EU.

In-class question

Can a strict preference for mixing two indifferent lotteries be explained by betweenness?

Weighted EU: Weak Independence

Replacement axiom: Weak Independence

If $P \sim Q$, then for each $\lambda \in (0, 1)$ there is a $\rho \in (0, 1)$ such that

$$\lambda P + (1 - \lambda)R \sim \rho Q + (1 - \rho)R \quad \text{for every } R.$$

The compensating probability ρ may differ from λ , but **must be the same for all common lotteries R** .

Chew and MacCrimmon's weighted EU representation is

$$V(P) = \frac{\sum_i p_i w(x_i) u(x_i)}{\sum_i p_i w(x_i)}, \quad w(x_i) > 0.$$

- Outcomes affect both utility and their effective probability weights.
- Constant w recovers EU; nonconstant weights allow Independence violations.

Chew and MacCrimmon (1979).

Weighted EU: Outcome Weights and Risk Attitudes

The effective probability of x_i is

$$p_i^w = \frac{p_i w(x_i)}{\sum_j p_j w(x_j)}.$$

- ▶ Larger $w(x_i)$ gives an outcome greater influence on evaluation.
- ▶ A useful interpretation: a prize is retained with probability $w(x_i)$; otherwise the lottery is redrawn. Normalize weights to at most one for this interpretation.
- ▶ Indifference lines can fan in or fan out, allowing Allais-type patterns for suitable utilities and weights. Betweenness still rules out a taste for randomization.

Behavioral implication for small risks

For smooth u, w and a small zero-mean risk ε ,

$$\text{risk premium} \approx \frac{1}{2} \left[-\frac{u''(x)}{u'(x)} - 2\frac{w'(x)}{w(x)} \right] \text{Var}(\varepsilon).$$

Decreasing outcome weights add to local risk aversion; increasing weights reduce it.

Implicit Utility: Very Weak Independence

Replacement axiom: Very Weak Independence

If $P \sim Q$, then for every R and $\lambda \in (0, 1)$ there exists $\rho(R) \in (0, 1)$ such that

$$\lambda P + (1 - \lambda)R \sim \rho(R)Q + [1 - \rho(R)]R.$$

Now the compensating probability **may depend on the common lottery R** .

Restriction	Required compensation
EU Independence	$\rho = \lambda$
Weighted EU weak independence	ρ independent of R
Implicit utility very weak independence	$\rho = \rho(R)$

This retains betweenness while allowing more flexible responses to common risk.

Implicit Weighted Utility and Implicit Expected Utility

Implicit weighted utility (Chew, 1985):

$$V(P) = \frac{\sum_i p_i w(x_i, V(P)) u(x_i)}{\sum_i p_i w(x_i, V(P))}, \quad w > 0.$$

Implicit expected utility (Dekel, 1986):

$$V(P) = \sum_i p_i \Gamma(x_i, V(P)).$$

- Solve for the lottery's value jointly with its outcome contributions. The representation requires conditions ensuring a well-defined value.
- With weights independent of V , implicit weighted utility reduces to WEU.

Behavioral implication

The same outcome can receive different weight in a good and a bad lottery. Indifference lines remain straight but need not be parallel; randomization between indifferent prospects remains neutral.

Disappointment Aversion: Restricted Independence

An outcome is disappointing if it is worse than the lottery's own certainty equivalent.

Replacement axiom: Gul's Weak Independence (intuition)

Mixing with a common sure outcome preserves rankings when the original lotteries have the same disappointment probability and the mixing does not switch original outcomes between disappointment and elation.

- ▶ This is a restriction on when ordinary mixing invariance applies; it differs from WEU's compensating-probability axiom.
- ▶ Together with ordering and continuity, it implies betweenness. An additional symmetry axiom gives Gul's particular representation.

Behavioral implication

A common consequence can change which outcomes disappoint. Then Independence need not hold, permitting a special attraction to certainty.

Gul (1991), Axioms 3–4. Betweenness alone does not characterize disappointment aversion.

Disappointment Aversion: Overweighting Bad Outcomes

For $\alpha \geq 0$, Gul's utility value solves

$$V(P) = \frac{\sum_i p_i w(x_i, V(P)) u(x_i)}{\sum_i p_i w(x_i, V(P))}, \quad w(x, V) = \begin{cases} 1, & u(x) \geq V, \\ 1 + \alpha, & u(x) < V. \end{cases}$$

Equivalently,

$$V(P) = \mathbb{E}_P[u(x)] - \alpha \mathbb{E}_P[(V(P) - u(x))_+], \quad (z)_+ = \max\{z, 0\}.$$

- ▶ α measures the extra penalty on disappointing outcomes.
- ▶ The reference level $V(P)$ is determined by the chosen lottery itself.
- ▶ $\alpha = 0$ recovers EU. Concave u and $\alpha > 0$ combine ordinary risk aversion with an additional disappointment penalty.

Disappointment Aversion and the Allais Choices

Set $u(x) = x$ and $\alpha = 1$. The DA values solve the preceding fixed-point equation.

	$p(2,500)$	$p(2,400)$	$p(0)$	EU value	DA value
<i>A</i>	.33	.66	.01	2,409	2,385.15
<i>B</i>	0	1	0	2,400	2,400.00
<i>C</i>	.33	0	.67	825	494.01
<i>D</i>	0	.34	.66	816	491.57

For these DA values, only the zero payoff is disappointing. For example,

$$V(A) = \frac{.33(2500) + .66(2400) + 2(.01)(0)}{.33 + .66 + 2(.01)} = \frac{2409}{1.01}.$$

Behavioral implication

EU with linear u predicts $A \succ B$ and $C \succ D$. Disappointment aversion instead gives $B \succ A$ and $C \succ D$.

Generalized Disappointment Aversion

Routledge and Zin (2010) allow disappointment to start below a fraction $\delta \in (0, 1]$ of the certainty equivalent $c(P)$ (positive monetary outcomes):

$$u(c(P)) = \mathbb{E}_P[u(x)] - \theta \mathbb{E}_P[(u(\delta c(P)) - u(x))_+], \quad \theta \geq 0.$$

Axiom change and nesting

The model remains in the Chew–Dekel betweenness class, with restrictions adapted to its disappointment threshold. Full Independence is not imposed. $\delta = 1$ gives Gul's model; $\theta = 0$ gives EU.

Behavioral implication

With $\delta < 1$, only sufficiently bad outcomes trigger the extra penalty. The strength of disappointment (θ) and its onset (δ) are separate: this can concentrate aversion on lower-tail outcomes.

Routledge and Zin (2010), Internet Appendix, equation (A1).

Local EU: Smoothness Instead of Global Independence

Machina (1982) asks which EU tools survive when Independence fails.

Organizing assumption: Differentiability of V

Retain a complete, transitive, continuous ordering and assume a smooth lottery value.

$$V(P + \Delta P) - V(P) = \sum_i \Upsilon(x_i; P) \Delta p_i + o(\|\Delta P\|), \quad \sum_i \Delta p_i = 0.$$

- ▶ $\Upsilon(x_i; P)$ is the **local utility** at the current lottery P .
- ▶ Infinitesimal probability changes are evaluated like EU, but local utility may change when the background lottery changes.
- ▶ The expansion is an approximation, not exact Independence for finite mixtures.

Behavioral implication

Indifference curves can bend. Risk attitudes and probability tradeoffs can depend on the current lottery, allowing common-consequence reversals.

Local EU: A Nonlinear Example

Consider

$$V(P) = \sum_i p_i u(x_i) + \frac{\eta}{2} \left[\sum_i p_i v(x_i) \right]^2.$$

Its local utility is

$$\Upsilon(x_i; P) = u(x_i) + \eta \left[\sum_j p_j v(x_j) \right] v(x_i).$$

The approximation error is quadratic: $O(\|\Delta P\|^2)$.

- ▶ For $\eta = 0$, EU is recovered. Otherwise, the background lottery generally changes local tradeoffs.
- ▶ Curvature alone need not change choices if V is merely an increasing transformation of an EU index.
- ▶ Increasing, concave local utilities at every lottery preserve FOSD and SOSD rankings under the smoothness conditions.

Cautious EU: Negative Certainty Independence

Let δ_x denote the lottery paying x with certainty.

Replacement axiom: Negative Certainty Independence (NCI)

For every P, R, x and $\lambda \in [0, 1]$,

$$P \succeq \delta_x \implies \lambda P + (1 - \lambda)R \succeq \lambda \delta_x + (1 - \lambda)R.$$

- ▶ If a risky prospect beats a sure outcome, adding common risk cannot reverse that ranking.
- ▶ The reverse comparison is unrestricted: a preference for certainty may disappear when both options become risky.

Behavioral implication: the certainty effect

$\delta_x \succ P$ can coexist with $\lambda P + (1 - \lambda)R \succ \lambda \delta_x + (1 - \lambda)R$. Removing certainty can reverse the choice.

Cerreia-Vioglio, Dillenberger, and Ortoleva (2015); NCI: Dillenberger (2010).

Cautious EU: The Lowest Certainty Equivalent

For each strictly increasing, continuous utility function $u \in \mathcal{U}$, calculate

$$c(P, u) = u^{-1} \left(\sum_i p_i u(x_i) \right), \quad V(P) = \inf_{u \in \mathcal{U}} c(P, u).$$

Use a minimum when it is attained. Rank lotteries by $V(P)$.

- ▶ The agent considers several possible utility functions and uses the **lowest monetary certainty equivalent**.
- ▶ For a sure outcome, all evaluations agree: $c(\delta_x, u) = x$.
- ▶ A singleton $\mathcal{U} = \{u\}$ recovers EU. Enlarging $\mathcal{U}$ can only lower $V(P)$, leaving sure outcomes unchanged.

Risk attitude and the axiom

If every $u \in \mathcal{U}$ is concave, then $V(P) \leq \mathbb{E}_P[x]$. If $P \succeq \delta_x$, every u ranks P above x ; each EU comparison survives mixing, which gives NCI after taking the infimum.

Which Restriction Replaces Independence?

Model	Main replacement or organizing restriction
Chew–Dekel	Betweenness: mixtures lie between their components.
Weighted EU	Weak Independence: compensation ρ can differ from λ , but is common across R .
Implicit utility	Very Weak Independence: compensation $\rho(R)$ can depend on R .
Disappointment aversion	Gul's restricted Weak Independence plus symmetry; betweenness is retained.
Local EU	Smoothness and local linear approximation; no global mixing axiom.
Cautious EU	Negative Certainty Independence: one-sided mixing invariance against a sure outcome.

Behavioral Predictions and Limits

Model	Distinctive behavioral implication
Betweenness family	Allais-type reversals are possible, but mixing indifferent lotteries cannot be strictly desirable.
Weighted / implicit utility	Outcome weights change effective risk attitudes; implicit weights also respond to overall lottery value.
Disappointment aversion	Avoid outcomes below an endogenous threshold; generalized DA can focus on the lower tail.
Local EU	Risk tradeoffs vary with the current lottery while familiar local risk-analysis tools remain useful.
Cautious EU	Certainty can be preferred before common risk is added, with the ranking reversing afterward.

References for Alternatives to EU

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Early Models of Reference Dependence

- ▶ Markowitz (1952) first proposed defining utility in terms of gains and losses relative to current wealth to explain why at all wealth levels individuals might both purchase insurance and take gambles.
- ▶ Kahneman and Tversky (1979) use this idea, together with probability weighting, to explain Allais and the other puzzles for expected-utility discussed before.
- ▶ Prospect theory takes the reference point as given. We will discuss some forward-looking models of endogenous reference point formation later.

Prospect Theory

Kahneman and Tversky (1979) describe a formula for evaluating simple prospects (gambles) of the form $(x, p; y, q)$ where $p + q \leq 1$. If $p + q < 1$, the remaining probability is understood to be assigned to zero.

Two ingredients in the model:

1. Value function $v(x)$, which satisfies $v(0) = 0$.
2. Weighting function $\pi(p)$, which satisfies $\pi(0) = 0$ and $\pi(1) = 1$.

Evaluation of lotteries (assume $x > y$):

$$V(x, 1) = V(x) = v(x)$$

$$V(x, p; y, q) = \pi(p)v(x) + \pi(q)v(y) \quad \text{if } p + q < 1 \text{ or } x \geq 0 \geq y$$

$$V(x, p; y, q) = v(y) + \pi(p)[v(x) - v(y)] \quad \text{if } p + q = 1 \text{ and } x > y > 0$$

Shape of the Value Function

Aversion to symmetric bets of the form $(x, 0.5; -x, 0.5)$ also implies

$$0 = v(0) > \pi(0.5)v(x) + \pi(0.5)v(-x) \Leftrightarrow v(x) < -v(-x).$$

- Significant concavity around the reference point 0 (e.g., kinks) can help explain scale-of-risk related paradoxes for EU (Rabin paradox, equity premium puzzle, etc.).

Experiment

KT (1979)

PROBLEM 13: Choose between

$$A : \begin{cases} 6,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.75 \end{cases} \quad \text{and} \quad C : \begin{cases} 4,000 & \text{with probability } 0.25 \\ 2,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.5 \end{cases}$$

$$\begin{aligned} B \succ A &\implies \pi(0.25)v(6000) < \pi(0.25)v(4000) + \pi(0.25)v(2000) \\ &\implies v(6000) < v(4000) + v(2000) \implies \text{concave for gains} \end{aligned}$$

PROBLEM 13': Choose between

$$C : \begin{cases} -6,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.75 \end{cases} \quad \text{and} \quad D : \begin{cases} -4,000 & \text{with probability } 0.25 \\ -2,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.5 \end{cases}$$

Recap: What Can Be Explained So Far

- ▶ We observed that significant concavity in the value function around the reference point might help to explain the scale of risk puzzles for expected utility:
 - ▶ Rabin paradox
 - ▶ Equity premium puzzle (see, Benartzi and Thaler (1995))
 - ▶ Kinks at 0 generate first-order risk aversion (as opposed to second-order risk aversion for expected utility)
- ▶ But we cannot yet explain the Allais paradox: Fixing the reference point at 0, the value function v is just an expected utility function and hence is subject to the paradox unless we introduce probability weighting.

Probability Weighting

- ▶ Kahneman and Tversky (1979) show that explaining the Allais common consequence effect requires $\pi(0.66) + \pi(0.34) < 1$.
- ▶ They also argue that risk taking for low-probability implies overweighting of low probability events.

One problem for the KT weighting function is sensitivity to framing.

- ▶ For example suppose $2\pi(p) > \pi(2p)$ for some $p < 1/2$
- ▶ Then, for $x > 0$,

$$V(x, p; x, p; 0, 1 - 2p) = 2\pi(p)v(x) > \pi(2p)v(x) = V(x, 2p; 0, 1 - 2p).$$

- ▶ With examples like this in mind, KT proposed an **editing phase** where the framing of gambles is set: $(x, p; x, p; 0, 1 - 2p)$ must be treated like $(x, 2p; 0, 1 - 2p)$.

Problems with KT weighting Function

Unfortunately, imposing a framing requirement does not resolve the issue.

- ▶ Suppose that $2\pi(p) > \pi(2p)$, but alter the previous example slightly.
- ▶ Suppose $x > 0$. Then for $\epsilon > 0$ sufficiently small,

$$V(x, p; x-\epsilon, p; 0, 1-2p) = \pi(p)v(x) + \pi(p)v(x-\epsilon) > \pi(2p)v(x) = V(x, 2p; 0, 1-2p).$$

- ▶ This is a violation of stochastic dominance.
- ▶ In light of such examples, KT proposed that, in addition, stochastically dominated alternatives are dropped during the editing phase prior to applying the function V .

Rank-Dependent Utility

Definition

A **rank-dependent utility (RUD)** representation consists of :

1. a utility function $u : X \rightarrow \mathbb{R}$ for outcomes; and
2. a function $\varphi : [0, 1] \rightarrow [0, 1]$ (**probability distortion function**) that is continuous, nondecreasing, and onto.

The utility of a simple lottery of the form $P = (x_1, p_1; \dots; x_n, p_n)$ with outcomes ordered $x_1 < x_2 < \dots < x_n$ is

$$U(P) = \sum_{i=1}^n \pi_i u(x_i),$$

where

$$\pi_i = \varphi(p_1 + \dots + p_i) - \varphi(p_1 + \dots + p_{i-1})$$

for $i > 1$ and $\pi_1 = \varphi(p_1)$.

Illustration of Probability Distribution

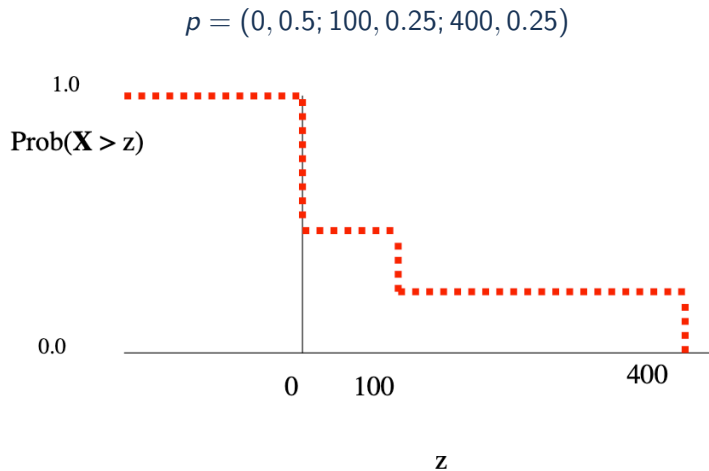


Figure: decumulative probability distribution

Illustration of EV

$$p = (0, 0.5; 100, 0.25; 400, 0.25)$$

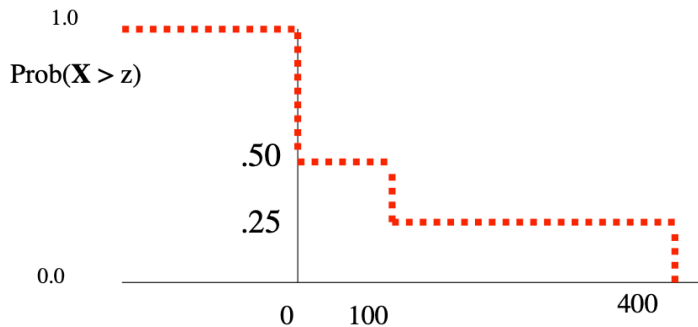


Figure: Expected Value

$$EV(p) = 0.5 \times 100 + 0.25 \times (400 - 100).$$

Illustration of EU

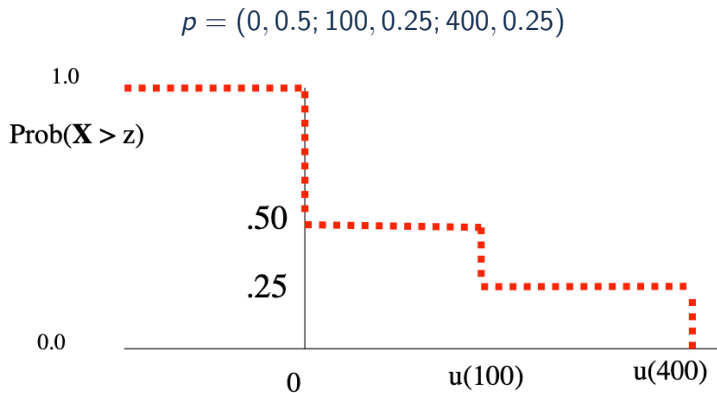


Figure: Expected Utility

$$EU(p) = 0.5 \times u(100) + 0.25 \times (u(400) - u(100)).$$

Illustration of RDU

$$p = (0, 0.5; 100, 0.25; 400, 0.25)$$

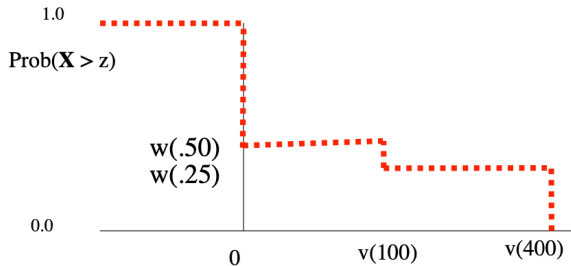


Figure: Rank-Dependent Utility

$$\begin{aligned} \text{RDU}(p) &= w(0.5) \times v(100) + w(0.25) \times (v(400) - v(100)) \\ &= (w(0.5) - w(0.25))v(100) + w(0.25) \times v(400) \\ &= [(1 - \varphi(0.5)) - (1 - \varphi(0.75))]v(100) + (1 - \varphi(0.75))v(400) \end{aligned}$$

Special Cases and Properties

- ▶ No probability distortion ($\varphi(p) = p$ identity function) is **expected utility**.
- ▶ Linear utility ($u(x) = x$) is the **dual theory** of Yaari (1987).
- ▶ Concavity of u and φ both play a role in risk aversion.

Example (binary lotteries, $u(x) = x$.)

$$\begin{aligned}U(x_1, p_1; x_2, p_2) &= \varphi(p_1)x_1 + (1 - \varphi(p_1))x_2 \\&< p_1x_1 + (1 - p_2)x_2 \quad (\text{if } \varphi(p_1) > p_1) \\&= U(p_1x_1 + p_2x_2, 1)\end{aligned}$$

Theorem

A RDU preference is risk averse iff both u and φ are concave.

Rank-Dependent Utility for General lottery

Definition

For non-simple lotteries, **rank-dependent utility** simply transforms the cdf F using the distortion function φ :

$$U(F) = \int u(x) d(\varphi \circ F)(x).$$

For the cdf of a simple lottery, this reduces to the previous formula.

- ▶ The formula above is the cumulative version of RDU (transformation of the cumulative distribution).
 - ▶ Some papers instead use the decumulative formula for RDA (transformation of the decumulative distribution).
 - ▶ The cumulative and decumulative versions of RDU are equivalent after suitable transformation of the probability distortion function.

RDU and the Allais Paradox

Probability	0.01	0.10	0.89
a_1	€1M	€1M	€1M
a_2	0	€5M	€1M
a_3	€1M	€1M	0
a_4	0	€5M	0

Suppose $u(x) = x$.

$$\begin{aligned}a_1 \succ a_2 &\Leftrightarrow 1\delta_{1M} \succ 0.01\delta_0 + 0.89\delta_{1M} + 0.1\delta_{5M} \\&\Leftrightarrow u(1M) > u(0)\varphi(0.01) + u(1M)[\varphi(0.9) - \varphi(0.01)] + u(5M)[1 - \varphi(0.9)] \\&\Leftrightarrow \frac{1 - \varphi(0.9) + \varphi(0.01)}{1 - \varphi(0.9)} > \frac{5M}{1M}\end{aligned}$$

$$\begin{aligned}a_3 \prec a_4 &\Leftrightarrow 0.89\delta_0 + 0.11\delta_{1M} \prec 0.9\delta_0 + 0.1\delta_{5M} \\&\Leftrightarrow u(0)\varphi(0.89) + u(1M)[1 - \varphi(0.89)] < u(0)\varphi(0.9) + [1 - \varphi(0.9)]u(5M) \\&\Leftrightarrow \frac{1 - \varphi(0.89)}{1 - \varphi(0.9)} < \frac{5M}{1M}\end{aligned}$$

Exercises

Exercises

1. Demonstrate that RDU can resolve the common consequence paradox.
2. Demonstrate that RDU can resolve the common ratio paradox.

RDU and the Rabin Paradox

Take $u(x) = x$ for simplicity

$$\begin{aligned}U(w - L, 0.5; w + G, 0.5) &\geq U(W) \\ \Leftrightarrow \phi(0.5)(w - L) + (1 - \phi(0.5))(w + G) &\geq w \\ \Leftrightarrow \frac{G}{L} &\geq \frac{\phi(0.5)}{1 - \phi(0.5)}\end{aligned}$$

Example

Let $\phi(0.5) = 3/5$, then accept lottery iff $G/L \geq 3/2$.

L	G	accept /reject
100	125	reject
10 000	16 000	accept

Is that for the Rabin Paradox?

Most people do not consider the Rabin paradox resolved because of the **background risk**.

- ▶ Suppose I offer you a gain 110 or loss 100 lottery.
- ▶ This is probably not the only risk you face today, and certainly not the only risk you face over a longer period such as a year.
- ▶ If we integrate this background risk $\tilde{y}$ into your decision about the lottery $\tilde{x}$, then you are really choosing between $\tilde{x} + \tilde{y} + w$ and $\tilde{y} + w$ rather than $\tilde{x} + w$ and w .
- ▶ It turns out that sufficiently large $\tilde{y}$ makes it difficult to generate aversion to the small-stakes Rabin lotteries, so the paradox remains.
- ▶ One possible solution is narrow framing.

Cumulative Prospect Theory

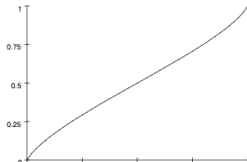
Tversky and Kahneman (1992) incorporate rank-dependent probability weighting into their model of prospect theory.

- ▶ They suggest decomposing any lottery into gains and losses and applying the rank-dependent formula to cumulative gain and loss probabilities separately.
- ▶ They suggest that a concave-convex shape for the weighting functions best fits the evidence (decumulative formula):

$$\varphi(p) = \frac{p^\beta}{(p^\beta + (1-p)^\beta)^{1/\beta}},$$

for some $\beta \in (0, 1)$.

- ▶ Overweights both low probability bad outcomes (certainty effect) and low probability good outcomes (possibility effect).



Loss Averse utility

- ▶ $u(x)$ is increasing in x , convex for loss, concave for gains, and first order concave at 0, that is

$$\lim_{x \rightarrow 0+} \frac{-u(-x)}{u(x)} = \lambda > 1.$$

- ▶ A useful parametrization

$$u(x) = \begin{cases} x^\beta & \text{for } x \geq 0 \\ -\lambda(-x)^\beta & \text{for } x \leq 0 \end{cases}$$

Exercise

Let $\beta = 1$. Characterize the range of β such that it can accommodate Rabin Paradox.

Second order risk aversion of EU

- ▶ Consider a lottery $(0.5, x + \sigma; 0.5, x - \sigma)$.
- ▶ Question: what risk premium π would you pay to avoid risk σ ?
- ▶ The risk premium r makes you be indifferent between

$$x \sim (0.5, x + \sigma + r(\sigma); 0.5, x - \sigma + r(\sigma)).$$

- ▶ Assume twice differentiable and take Taylor expansion.

$$\begin{aligned} u(x) &= u(x) + 1/2 u'(x) \cdot 2r(\sigma) + 1/4 u''(x) \cdot 2(\sigma^2 + r(\sigma)^2) + o(\sigma^2) \\ r(\sigma) &= -\frac{u''(x)}{u'(x)} \frac{\sigma^2 + r(\sigma)^2}{2} + o(\sigma^2) = \frac{\rho}{2} [\sigma^2 + r(\sigma)^2] + o(\sigma^2) \\ &\approx \frac{\rho}{2} \sigma^2 \end{aligned}$$

First order risk aversion of RDU

- ▶ Consider same lottery $(0.5, x + \sigma; 0.5, x - \sigma)$.
- ▶ Question: what risk premium π would you pay to avoid risk σ ?
- ▶ Assume

$$u(x) = \begin{cases} \frac{x^{1-\gamma}}{1-\gamma} & \text{for } x \geq 0 \\ -\lambda \frac{(-x)^{1-\gamma}}{1-\gamma} & \text{for } x \leq 0. \end{cases}$$

- ▶ The premium r at $x = 0$ satisfies

$$\begin{aligned} 0 &= 1/2(\sigma + r(\sigma))^{1-\gamma} + 1/2(-\lambda)(-\sigma + r(\sigma))^{1-\gamma} \\ r(\sigma) &= \frac{\lambda^{\frac{1}{1-\gamma}} - 1}{\lambda^{\frac{1}{1-\gamma}} + 1} \sigma = k\sigma \end{aligned}$$

RDU Optimal Portfolio

Given the probability weights φ , the optimization problem for a RDU investor is

$$\begin{aligned} \max_{x_s} \quad & \sum_{s=1}^n \pi_s u(x_s) \\ \text{s.t.} \quad & \sum_{s=1}^n x_s q_s = w \\ & x_s \leq x_{s+1}, \end{aligned} \tag{1}$$

where q_s is price at state s and w is wealth.

Solution

The Lagrangian for this problem is

$$\mathcal{L} = \sum_s \pi_s u(x_s) + \eta \left[w - \sum_{s=1}^n q_s x_s \right] + \sum_{s=1}^{n-1} \kappa_s (x_{s+1} - x_s). \quad (2)$$

FOC:

$$0 = \partial \mathcal{L} / \partial x_s = \pi_s u'(x_s) - \eta q_s - \kappa_s + \kappa_{s-1}$$

$$0 \leq \partial \mathcal{L} / \partial \kappa_s = x_{s+1} - x_s$$

$$0 = \kappa_s (x_{s+1} - x_s)$$

Solution

For concave utility, marginal utility is invertible

$$x_s^* = u'^{-1}\left(\frac{\eta q_s + \kappa_s - \kappa_{s-1}}{\pi_s}\right). \quad (3)$$

► $x_{s-1}^* < x_s^* < x_{s+1}^* \implies \kappa_s = \kappa_{s-1} = 0,$

$$x_s^* = u'^{-1}\left(\frac{\eta q_s}{\pi_s}\right).$$

► $x_{s'}^* = \dots = x_{s''}^*,$

$$u'^{-1}\left(\frac{\eta q_{s''}}{\pi_{s''}}\right) \leq x_s^* \leq u'^{-1}\left(\frac{\eta q_{s'}}{\pi_{s'}}\right).$$

Example

States	s_1	s_2	s_3
Probabilities	0.2	0.3	0.5
State Prices	0.3	0.3	0.4

State Price per Probability Unit (SPP):

$$\frac{q_1}{p_1} = 1.5 > \frac{q_2}{p_2} = 1 > \frac{q_3}{p_3} = 0.8.$$

π -SPP

Probability weighting function:

$$\varphi(p) = \frac{p^\beta}{(p^\beta + (1-p)^\beta)^{1/\beta}},$$

for some $\beta = 0.7$.

$$\pi_1 = \varphi(0.2) = 0.2560 \quad \pi_2 = \varphi(0.5) - \varphi(0.2) = 0.201 \quad \pi_3 = 1 - \varphi(0.5) = 0.5426.$$

So, for π -SPP:

$$\frac{q_2}{\pi_2} = \frac{0.3}{0.201} > \frac{q_1}{\pi_1} = \frac{0.3}{0.256} > \frac{q_3}{\pi_3} = \frac{0.4}{0.5426}.$$

Hence an optimal portfolio is $x^* = (0.575, 0.315, 1.837)$.

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Hence an optimal portfolio is $x^* = (0.575, 0.315, 1.837)$.

But, correct?

Check

If $x^* = (0.575, 0.315, 1.837)$ is correct, then $x_2 < x_1 < x_3$. The weighted probability should be

$$\pi_2 = \varphi(0.3) = 0.3281 \quad \pi_1 = \varphi(0.5) - \varphi(0.3) = 0.1293 \quad \pi_3 = 1 - \varphi(0.5) = 0.5426.$$

Therefore, π -SPP

$$\frac{q_1}{\pi_1} = \frac{0.3}{0.1293} > \frac{q_2}{\pi_2} = \frac{0.3}{0.3281} > \frac{q_3}{\pi_3} = \frac{0.4}{0.5426}.$$

Inconsistency: This cannot be an optimal solution.

Exercise

Combine state 1 and 2 and calculate the optimal allocation. Then, verify the solution.